

LAB 03 | Active Directory HelpDesk Access & Permissions

IAM Portfolio Project | Windows Server 2022 | Active Directory | SMB File Sharing

Project overview

Configured and troubleshot role-based access to a HelpDesk network share in a simulated Active Directory domain. The lab used a test employee account and an IT HelpDesk security group to examine authentication, network connectivity, and authorization.

Lab environment

Component	Lab configuration
Domain controller	JMC-DC01 (Windows Server 2022)
Active Directory domain	jmc-local
Test identity	David Williams (dwilliams)
Role group	GG-IT-HelpDesk
Shared resource	\\JMC-DC01\JMC-HelpDesk

Objectives

- Verify a domain user account and HelpDesk security group membership.
- Test access to a shared folder using a domain user identity.
- Troubleshoot sign-in restrictions, network path errors, and permission denials.
- Demonstrate how group-based permissions support least-privilege access.

Troubleshooting and verification

Sign-in restriction: A test login initially reported that the sign-on method was not allowed.

Network access: The share was initially unreachable; a network-path-not-found error was observed.

Authorization: An access-denied message appeared before permission adjustments.

Follow-up: After troubleshooting, a command returned “The command completed successfully.” This supports progress but does not, on its own, prove final end-user file access.

Security principles demonstrated

Role-based access control (RBAC); Active Directory security groups; SMB share access; authentication versus authorization; least privilege; structured troubleshooting.

Evidence and next verification

Original Lab 3 screenshots are not included in this version. Before describing share access as fully verified, add a screenshot showing David Williams successfully opening the HelpDesk share and, ideally, the relevant group/share permission settings.

Portfolio takeaway

This project documents practical IAM troubleshooting across identity, network access, and permissions in a Windows domain. The evidence is described conservatively to distinguish resolved command-level checks from final user access verification.